THE HONG KONG UNIVERSITY OF SCIENCE & TECHNOLOGY

ELEC 3130: Digital Image Processing

Homework 1

Due Date: 2024.02.28 14:59

Your answers should be typed, not handwritten. You should submit a Word file or a pdf file.

Submissions are to be made via Canvas. ***Late submission is not allowed (0 point)***

Submitted File Name Format: name_ID_Homework1

1. Suppose A=$\begin{bmatrix} 3 & 3 \\ 1 & 0 \end{bmatrix}$. (20 points)

a) Is A a Hermitian matrix? Show your reasons. (5 points)

$A^T_{conj} = \begin{bmatrix} 3 & 1 \\ 3 & 0 \end{bmatrix} \neq A$

➔No

b) Calculate A^{-1}. (5 points)

$A^{-1} = \frac{1}{3*0-1*3} * \begin{bmatrix} 0 & -3 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{1}{3} & -1 \end{bmatrix}$

c) What are the eigenvalues of A? (5 points)

$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 3 \\ 1 & 0-\lambda \end{vmatrix} = (3-\lambda)(-\lambda) - 3 = -3\lambda + \lambda^2 - 3$

$-3\lambda + \lambda^2 - 3 = 0$

$\lambda_1 = \frac{-\sqrt{21}+3}{2}, \ \lambda_2 = \frac{\sqrt{21}-3}{2}$

d) Now regard the first column of A as vector x. Given rotation angle $\theta = 60°$, use a rotation matrix $\begin{bmatrix} cos\theta & sin\theta \\ -sin\theta & cos\theta \end{bmatrix}$ to rotate this vector x to a new one x'. Compute the new vector x'. (5 points)

$x = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

Sub $\theta = 60°$

R = $\begin{bmatrix} cos60° & sin60° \\ -sin60° & cos60° \end{bmatrix}$

x'=Rx=$\begin{bmatrix} cos60° & sin60° \\ -sin60° & cos60° \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{3+\sqrt{3}}{2} \\ \frac{1-3\sqrt{3}}{2} \end{bmatrix}$

2. Given a signal A= [2,4,6,7,8] and a filter B= [1, 0, 0.5]. (20 points)

Calculate the linear convolution of A and B.

A = [2, 4, 6, 7, 8]

B = [1, 0, 0.5]

$$y[n] = \sum_{k=0}^{n-1} x[k]h[n-k]$$

n=0 ➔ 2×1 = 2

n=1 ➔ 2×0 + 4×1 = 4

n=2 ➔ 2×0.5 + 4×0 + 6×1 = 7

n=3 ➔ 4×0.5 + 6×0 + 7×1 = 9

n=4 ➔ 6×0.5 + 7×0 + 8×1 = 11
n=5 ➔ 7×0.5 + 8×0 = 3.5
n=6 ➔ 8×0.5 = 4
y[n] = [2, 4, 7, 9, 11, 3.5, 4]

3.A hospital uses two backup generators, X and Y, during a power outage. The probability that generator X functions is 0.95, and the probability that generator Y functions is 0.88. The probability that both generators function simultaneously is 0.85. (20 points)

a) What is the probability that at least one generator functions during a power outage? (10 points)

$P(X \cup Y) = P(X) + P(Y) - P(X \cap Y)$
$= 0.95 + 0.88 - 0.85$
$= 0.98$

b) What is the probability that neither generator functions? (10 points)

P(neither generator functions) = $1 - P(X \cup Y)$
$= 1 - 0.98$
$= 0.02$

4. Let the probability density of the continuous random variable X be

$$f(x) = \begin{cases} \dfrac{2}{\pi\sqrt{1-x^2}} & , \quad 0 < x < c \\ 0 & , \quad others \end{cases}$$

a) What is the value of c? (10 points)

The integral of the probability density function (PDF) over its entire domain must equal 1, as this ensures that the total probability of all possible outcomes is normalized to unity.

so $\int_{-\infty}^{\infty} f(x)dx = 1$

$\int_0^c \frac{2}{\pi\sqrt{1-x^2}} dx = 1$

$\frac{2}{\pi}[\arcsin(x)]_0^c = 1$

$\frac{2}{\pi}\arcsin(c) = 1$

$\arcsin(c) = \frac{\pi}{2}$

so c = 1

b) Please find the X's distribution function F(x). (10 points)

F(x) = $\int_{-\infty}^{x} f(t)dt$

F(x) = $\int_{-\infty}^{x} \frac{2}{\pi\sqrt{1-t^2}} dt$

For x ≤ 0:

F(x) = $\int_{-\infty}^{x} 0\, dt = 0$

For 0 < x < 1:

$$F(x) = \int_0^x \frac{2}{\pi\sqrt{1-t^2}}dt$$

$$= \frac{2}{\pi}[\arcsin(t)]_0^x$$

$$= \frac{2}{\pi}\arcsin(x)$$

For x ≥ 1 :

$$F(x) = \int_0^1 \frac{2}{\pi\sqrt{1-t^2}}dt$$

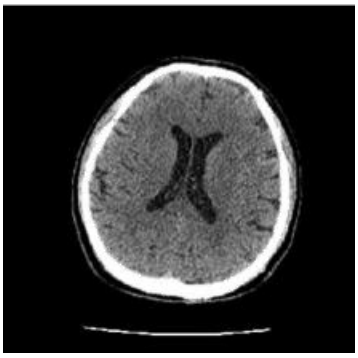
$$= \frac{2}{\pi}[\arcsin(t)]_0^1$$

$$= \frac{\pi}{2} * \frac{2}{\pi}$$

$$=1$$

$$F(x) = \begin{cases} \frac{2}{\pi}\arcsin(x), & \begin{matrix} 0, & x \leq 0 \\ 0 < x < 1 \\ 1, & x \geq 1 \end{matrix} \end{cases}$$

5. Use MATLAB to crop this image and show the results in the format of subplot. Here are three requirements: (20 points)



(1) In the first subplot, show the original image; (10 points)
(2) In the second subplot, crop any area of the image and show the result; (10 points)
(3) You should put your code and results in your submitted file.

hw1.mlx

```
clear; clc;

originalImage = imread('original_image.png');

figure(1);

subplot(1, 2, 1);
imshow(originalImage);
title('Original Image');

cropRect = [37, 32, 110, 110];

croppedImage = imcrop(originalImage, cropRect);

subplot(1, 2, 2);
imshow(croppedImage);
title('Cropped Image');
```

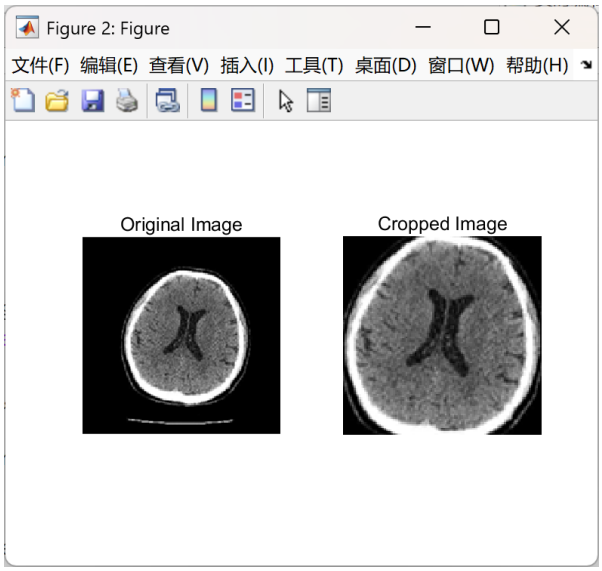


Figure 2: Figure
文件(F) 编辑(E) 查看(V) 插入(I) 工具(T) 桌面(D) 窗口(W) 帮助(H)
Original Image
Cropped Image